

General and local anesthetics

By Yitagesu T.(MSc.), 2024, AAU

Local anesthetics

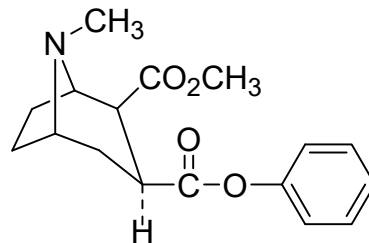
Local anesthetics

- **Block the nerve** that carries the **pain sensation** and **automatic impulses** in local areas of the body
- Prevent conduction and formation of an action potential by **either full or partial blockage** of **sodium ion channel**
- Used in **dentistry, ophthalmology** and **minor surgical operations**

Local ...

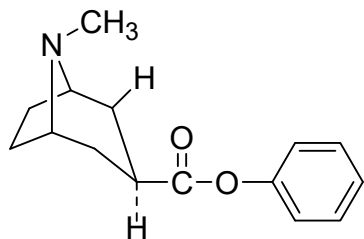
Ester and amide type

- Cocaine an alkaloid from the leaves of *Erythroxylon coca* was the first local anesthetic
- The development of local anesthetics began after the discovery of local anesthetic properties of cocaine

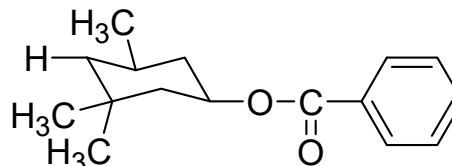


- Because of its **addictive** properties the search for non-habit forming local anesthetics began

Local ...

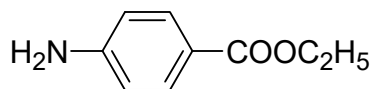


Tropacocaine



β -eucaine

- Based on the above findings many other simple analogs like procain and benzocaine was synthesized



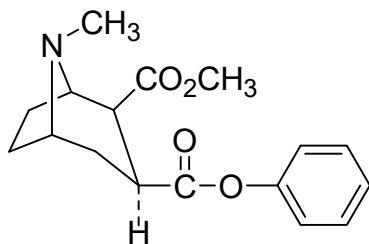
Benzocaine



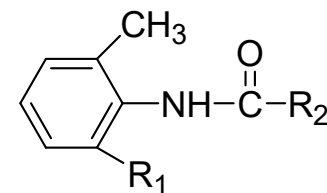
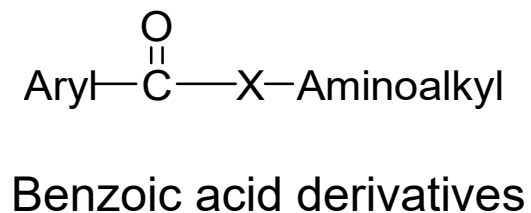
Procaine

Local ...

- General structural features



Cocaine



Anilides derivatives

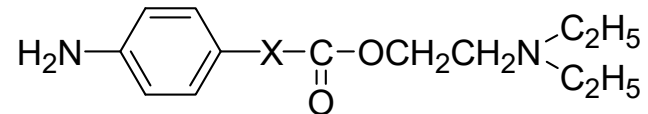
Local ...

SAR of local anesthetics

- The **lipophilic center** is usually either a cyclic or heterocyclic system
- The **hydrophilic center** is normally a secondary or tertiary amine, which may or may not be cyclic
 - Tertiary amines are more useful since they are less irritating to tissue
- The **hydrophilic center** may be attached to ester or amide by a short hydrocarbon chain

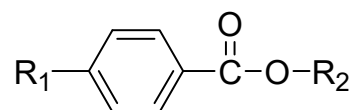
Local ...

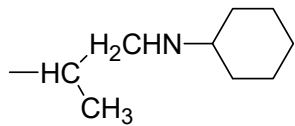
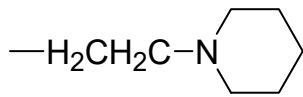
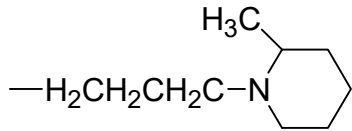
- The **lipophilic center** is responsible for **lipid solubility** as it affects cell penetration and their action
- The **hydrophilic center** provides **water solubility** and is believed to be involved in binding to the receptor
- For best action a balance between lipophilic and hydrophilic center is essential
- Ester type require a carbonyl group in conjunction with an aromatic ring or related system



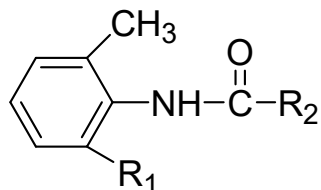
Local ...

- Benzoic acid derivatives**



Generic name	R1	R2
Hexylcaine		
dyclonine	CH ₃ CH ₂ CH ₂ CH ₂ O—	
piperocaine		

- Lidocaine (the amino amide)



Generic name	R1	R2
Etidocaine	CH3	
Mepivacaine	CH3	
Bupivacaine	CH3	
lidocaine	CH3	
prilocaine	CH3	

Yitagesu T(MSc.)

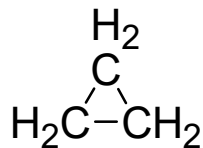
General Anaesthetics

- General anaesthesia is a controlled **reversible depression of the functional activity of the CNS**
- Early agents were **ether, chloroform & nitrous oxide**
- Nowadays, multiple drug regimes used including
 - Pre-anaesthesia + skeletal muscle relaxants + drugs to control side effects + actual anaesthetic agent are used in combination
 - General anaesthetics are given either by **inhalation** or i.v.

General...

- **Inhalation anaesthetics (gases and liquids)**
- Current inhalation drugs include halothane [CHBrClCF_3] & several other F&Cl-containing molecules, and nitrous oxide.

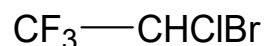
Gas inhalation anaesthetics



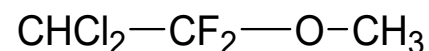
Cyclopropane



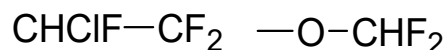
Nitrous oxide



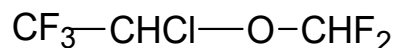
Halothane



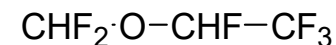
Methoxyflurane



Enflurane



Isoflurane



Desflurane

General ...

Structure activity relationships

- General anaesthetics are structurally **non-specific** and have the following characteristics
 - They are relatively **inert chemically** and do not directly interact with functional groups on many functional biopolymers
 - Their effective concentrations are generally higher than those of receptor bound drugs
 - The **more soluble** the drug in oil, the low dose is needed to produce the standard response because of its higher ability to reach the site of action

General ...

- **Intravenous anaesthetic**
 - They produce unconsciousness but not sufficient depth to permit surgical procedures
 - The **intravenous anaesthetics** is given prior to **inhalation anaesthetics**
- **Classification**
 - Ultrashort acting barbiturates
 - Benzodiazepines
 - Ketamine hydrochloride

